

Data Structure and Programming, Spring 2025
programming assignment #5
**Top-k Profitable Trading Periods in
Stock Returns**

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RELEASE DATE: 05/22/2025
DUE DATE: 06/05/2025

Late policy: $Score = Score \times 0.7^{\lceil h \rceil / 24}$, $h = \text{late hours} \leq 48$

1 Problem Description

You are a short-term trader analyzing a stock's performance over the past n days. Each day's profit or loss is recorded as an integer: positive for gains, negative for losses.

You are not just interested in the best single streak, but in understanding the top k most profitable trading streaks. A streak is defined as a contiguous range of days, and its "profit" is the sum of returns over that range.

Your goal is to identify **the top-k highest profits** that could have been made by holding the stock over any contiguous sequence of days. This will help you study past momentum and volatility patterns. For example, given daily returns: $[3, 1, -1]$. All possible contiguous streaks:

$[3]$: sum = 3, $[1]$: sum = 1, $[-1]$: sum = -1,
 $[3, 1]$: sum = 4, $[1, -1]$: sum = 0, $[3, 1, -1]$: sum = 3,

- Top-1 sum of subarray = $[4]$. (subarray = $[3, 1]$)



- Top-2 sum of subarray = [4, 3]. (subarray = [3, 1], [3] or [3, 1], [3, 1, -1])
- Top-3 sum of subarray = [4, 3, 3]. (subarray = [3, 1], [3], [3, 1, -1])

2 Implementation Requirements

We provide the template: **main.py** for input/output handling. Your task is to complete the sections marked as TODO.

- The input of the function is an array of length n ($arr[0, \dots, n-1]$) and an integer k .
- Each element in the array can be a positive or negative number.
- Return the the top- k highest subarray sums as a list.
- The time complexity of your code should not exceed $O(n^2 \log k)$

Note:

1. You are free to use any algorithm you prefer, e.g. brute-force, divide and conquer, heap data structure, etc. However, since we have set time limit constraint, it is recommended to consider efficiency of your method.
2. Do **NOT** modify the interface of the functions, but you can add your own functions.
3. All python standard libraries (except heapq) are allowed. Other libraries such as numpy are NOT allowed. If you have any question regarding library permission, you are welcome to email TA.
4. Environment: Python 3.10, Ubuntu 20.04.

3 Evaluation

We provide 3 testcases: input_1.json, input_2.json, input_3.json, and their corresponding golden outputs: golden_1.json, golden_2.json, golden_3.json. Use command “python3 main.py --input input_x.json --output output_x.json” ($x = 1, 2, 3$) to verify the correctness of your code. Comparing your result with golden_1.json, golden_2.json, and golden_3.json to check your code.

4 Submission

Please put `main.py` into a directory named `studentID` and compress the directory into `studentID.zip` to NTU COOL. The homework is due on 06/05, at 23:59.

e.g.

```
b10901154.zip/  
b10901154/  
main.py
```

5 Grading policy

This programming assignment will be graded based on correctness and time complexity.

- We have 5 test cases in total, and we provide 3 of these test cases. The total point is 100.
- If your program passes the test cases, for each case you will get 10%.
- If your program passes the test cases with desired time complexity, for each case you will get additional 10%.

Note: Incorrect submission format will result in a 20% penalty.